

MUGHAL ARSHAD

AI/ML Engineer · LLM & Generative AI · Full-Stack Developer · Automation
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PROFESSIONAL SUMMARY

Results-driven Computer Science engineer with hands-on experience building LLM-powered applications, AI pipelines, and automation systems. Proven track record of deploying Generative AI into production HRMS environments, developing NLP-based behavioural analysis engines, and applying machine learning to clinical datasets. Proficient in Python, FastAPI, LangChain-style agentic workflows, RAG pipelines, and full-stack development. Passionate about building reliable, explainable, and production-grade AI solutions.

CORE TECHNICAL SKILLS

LLMs & GenAI	Zephyr 7B Beta, OpenAI API, Prompt Engineering, RAG Pipelines, Agentic Workflows, LangChain, Guardrails
NLP & AI	Sentiment Analysis, Big Five Personality Modeling, NER, Document Understanding, Transformers, Scikit-learn
Python Stack	FastAPI, Pydantic, Selenium WebDriver, Flask, Pandas, Pygame — PEP-8 clean, production-tested
Document AI	PDF/DOCX Parsing, Resume Scoring Engine, OCR Integration, Automated PDF Report Generation
Frontend	React.js, TypeScript, Vite, JavaScript, HTML5, CSS3
Data & BI	Power BI, SQL, MySQL, Exploratory Data Analytics, Predictive Modeling, Statistical Analysis
Infra & DevOps	JWT Auth, RBAC, REST API Design, Swagger/ReDoc, Git, CI-friendly testing with Selenium

INTERNSHIP & WORK EXPERIENCE

Product & Engineering Intern

2025 – Present

Vivnovation

- Engineered Selenium + Python automated test pipelines that reduced manual QA effort by ~40%, accelerating release cycles across the engineering team.
- Delivered 3+ software features end-to-end within a cross-functional product team, contributing from requirements through deployment.
- Identified and automated 10+ repetitive QA workflows, improving test coverage consistency and reducing regression discovery time.
- Collaborated with senior engineers on full-stack development tasks, applying Python and web technologies to production-grade systems.

Software Engineer — AI & ML

Jun 2025 – Aug 2025

Brilyant IT Solutions

- Architected and deployed an LLM-powered Leave Processor using Zephyr 7B Beta, reducing HR processing time by automating 100% of routine leave approvals through natural language understanding.
- Engineered a policy-compliance validation engine that cross-referenced employee leave requests against HR rule sets, eliminating manual eligibility checks.
- Integrated Generative AI into a production HRMS application, delivering the first AI-driven workflow automation module for enterprise HR operations.
- Designed prompt engineering strategies and fallback logic to ensure robust, policy-compliant outputs under varied input quality.

Data Analyst

Feb 2025 – May 2025

Technocolabs Softwares

- Processed and analyzed large-scale logistics datasets using Python and SQL, uncovering optimization opportunities for AI & GenAI-based routing solutions.
- Developed 4+ interactive Power BI dashboards adopted by cross-functional stakeholders for data-driven decision-making.
- Collaborated on AI pipeline design for demand forecasting, applying statistical modeling and EDA techniques to real operational data.

Clinical AI Intern

May 2024 – Jun 2024

The Apollo Hospitals

- Led a 4-member team to build a stroke prediction model using ML classification algorithms, achieving measurable accuracy improvements on patient datasets.
- Applied EDA and feature engineering to clinical patient data, identifying 5+ key risk factors that informed the predictive model architecture.
- Evaluated model performance using precision, recall, and F1 metrics — gaining foundational experience in responsible AI and model monitoring.

KEY PROJECTS

AI-Powered Recruitment Platform (Full-Stack) · Python · FastAPI · LLMs · NLP · React · TypeScript · JWT · RAG

- Architected a production-grade AI recruitment platform with a 7-module FastAPI backend, processing multi-stage candidate evaluations from resume ingestion to final scoring.
- Built an LLM-driven behavioural analysis engine generating role-specific questions and scoring responses across Sentiment, Big Five Personality (5 dimensions), and 5 Soft-Skill indicators.
- Engineered a resume parsing pipeline that extracts skills from PDF/DOCX files, matches against job requirements, and generates weighted scores (0–100) — reducing manual screening time by up to 70%.
- Implemented a Combined Scoring Engine (50/50 resume + behavioural) with Fit Score ranking, enabling recruiters to shortlist top candidates in under 30 seconds.
- Deployed JWT-based authentication with RBAC, protecting 15+ API endpoints across distinct Candidate and Recruiter roles with full ownership enforcement.
- Incorporated RAG-style fallback processing for AI pipeline resilience, maintaining system uptime and scoring accuracy under LLM unavailability scenarios.

AI-Based Leave Processor — HRMS · Python · Zephyr 7B Beta · LLMs · Generative AI · NLP · FastAPI

- Deployed a Zephyr 7B Beta LLM pipeline that processes natural language leave requests, automating eligibility validation and approval decisions with 0 manual HR touchpoints for routine cases.
- Designed agentic workflow with multi-step reasoning: request parsing → policy lookup → eligibility check → approval/rejection — reducing average processing time from hours to seconds.

Student Management System · Python · Flask · JavaScript · HTML · CSS · MySQL

- Developed a full-stack Flask web application managing 3 core data domains — personal records, attendance, and grades — with MySQL relational backend and dynamic frontend.

EDUCATION

B.Tech — Computer Science Engineering

Expected May 2026

The Apollo University, Chittoor · CGPA: 9.47 / 10.0

LEADERSHIP & ACTIVITIES

- President, Computer Society of India — Student Chapter, The Apollo University: Led the chapter, organized tech events, and mentored peers in programming and AI.
- Student Coordinator, RWTH University & The Apollo University: Planned and hosted 10+ academic and non-academic events, driving cross-institutional collaboration.
- Member, The Apollo Model UN Club: Developed communication, negotiation, and research skills through competitive MUN conferences.